

MICROFI

UI/UX DESIGN HANDOFF

Version 3.0 — Aligned to System Architecture & Software Design (SAD) v1.1 **and the live Core Backend API**
August 2026

Application: Digital Cash Collection Network — CEMAC Region
Targets: Flutter Mobile App (Agent + Client) • Next.js 16 Back-Office

Traceable to FR-01...FR-22, NFR-01...NFR-12, BR-01...BR-05, UC-01...UC-22

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0. Purpose, Scope & Traceability

This handoff translates the MICROFI System Architecture and Software Design (SAD v1.1) into concrete screen-level UI specifications for three surfaces: the Flutter Agent app, the Flutter Client app, and the Next.js 16 Back-Office. Every screen in this document is traceable to a use case (UC), functional requirement (FR), non-functional requirement (NFR) or business rule (BR) defined in the SAD and its upstream Business Analysis.

This document supersedes v2.0 of the UI/UX handoff. v2.0 was cross-checked directly against the running Core Backend API (not just the SAD document), and several sections were found to describe a flow the API does not implement. Corrections in this v3.0 revision:

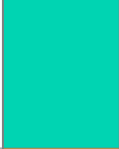
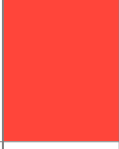
- **UC-19 activation is a two-party, two-device flow, not a single agent-side entry.** The client sets their own credentials and confirms payment on their *own* device, in two separate API calls, not by handing their Activation ID to the agent to type in. This was a deliberate anti-fraud design decision (see §5.1) — the earlier single-actor flow would let an agent activate an account and collect commission without genuine client involvement.
- **The digital booklet token has three real states (ACTIVE / EXPIRED / REVOKED), not four.** There is no GRACE state in the backend today — see §5.2.
- **The daily CBS export does not go through RabbitMQ.** It is a direct synchronous call from Core to the Middleware, protected by an idempotency key, not an async queue — see §8.1.
- The PIN-lockout (§4.1) and OFJ-close-blocked-on-pending-sync (§6.1) behaviors described in v2.0 did *not* yet exist in the backend at the time of that review. Rather than soften the spec, both were implemented server-side to match this document — see the refined mechanism notes in each section.

See the Glossary (final section) for definitions of all acronyms and domain terms used throughout this document.

1. Design Tokens

1.1 Color System

Token	Color name	Swatch	Hex	Usage	Data binding

color/primary/ navy	Deep Navy Blue		#0A2540	Brand, headers, primary actions	—
color/success/ emerald	Emerald Green		#00D4B2	Synced, validated, success CTA	sync_status = SYNCED
color/warning/ amber	Amber		#FFB800	Offline, pending, ≥80% ceiling	sync_status = PENDING
color/danger/red	Signal Red		#FF453A	Blocked, SOS, 100% ceiling, GPS fail	escrow ceiling reached
color/surface/ white	White		#FFFFFF	App/card backgrounds	—
color/surface/grey-100	Light Grey		#F2F2F2	Section backgrounds, dividers	—
color/text/slate	Slate Grey		#334155	Secondary text, captions	—
color/text/grey-disabled	Cool Grey		#9CA3AF	Disabled labels, expired states	access_token.status = EXPIRED/REVOKED

| Palette logic: a navy-and-neutral base (Deep Navy Blue, White, Light Grey, Slate Grey, Cool Grey)
 | provides structure, while three traffic-light accents — Emerald Green, Amber, Signal Red — carry state
 | (good / caution / stop). Amber is always paired with black text, never white, to hold contrast in direct
 | sunlight. Status is never conveyed by color alone: every colored state pairs a color with an icon and a text
 | label.

1.2 Typography Scale

Style	Size / Weight	Mobile font	Web font	Usage
Display	32 pt / Bold	Inter Bold	Roboto Bold	Passbook balance, OFJ totals

H1	22 pt / Bold	Inter Bold	Roboto Bold	Screen titles
H2	18 pt / SemiBold	Inter SemiBold	Roboto Medium	Section headers
Body	15 pt / Regular	Inter Regular	Roboto Regular	Default reading text
Caption	12 pt / Regular	Inter Regular	Roboto Regular	Helper text, timestamps
Numeric/ Mono	15 pt / Tabular	Inter Tabular Nums	Roboto Mono (receipts)	XAF amounts, denomination grid

Amount display rule (unchanged from BA): the currency code never precedes the amount. Strict format: [amount][non-breaking space]XAF, e.g. 45 000 XAF.

1.3 Spacing, Radius & Elevation

Token	Value	Usage
space-1	4 dp/px	Icon-to-label gap
space-2	8 dp/px	Compact list padding
space-3	12 dp/px	Default field padding
space-4	16 dp/px	Card padding, section gaps
space-6	24 dp/px	Screen margins (mobile)
space-8	32 dp/px	Section separation (web)
radius-sm	6 px	Inputs, chips
radius-md	12 px	Cards, modals

radius-full	999 px	Pills, status badges, FAB
elevation-1	0 1px 2px rgba(0,0,0,.08)	Cards at rest
elevation-2	0 4px 12px rgba(0,0,0,.12)	Modals, floating action button

1.4 Responsive Breakpoints (Next.js Back-Office)

Breakpoint	Width	Back-Office (Next.js) layout
sm	< 640 px	Not supported for agent admin flows; redirect to mobile-friendly notice
md	640–1024 px	Single-column dashboard, collapsible side nav, stacked map/table
lg	1024–1440 px	Two-column dashboard, persistent side nav, side-by-side map/table
xl	> 1440 px	Full grid: nav + content + detail panel (agent drill-down)

2. Component Library & Interaction States

Every reusable component must ship all states below — not just the default appearance shown in mockups — so engineering can implement without guessing intermediate states.

2.1 Buttons

State	Visual	Trigger	FR/UC reference
Default	Navy fill, white text, radius-sm, 48 dp height min.	Ready for input	Fat-finger spec, all wizards
Disabled	Grey #E5E7EB fill, grey-disabled text	Preconditions unmet (e.g. denomination sum $\neq$ amount, GPS off, justification <	BR-02, BR-05, UC-17

		10 chars)	
Loading	Navy fill, inline spinner replaces label, button non-interactive	Awaiting server/CBS response	NFR-05, UC-09
Success	Emerald fill, checkmark icon, auto-reverts after 2s or on navigation	Delta == 0 close, collection validated	UC-16, UC-07
Danger/ Destructive	Red outline or fill, confirmation required before commit	Suspend agent, write off variance	UC-02, UC-17

2.2 Input Fields

State	Visual	Notes
Default	1px grey-300 border, radius-sm, 48 dp height	Numeric fields restrict to integers (denomination grid, FR-08)
Focus	2px navy border, no shadow (avoid outer glow that reduces sunlight contrast)	Focus ring must remain visible under direct sunlight
Error	2px red border + red helper text below field	e.g. denomination sum mismatch (BR-02), justification < 10 chars (UC-17)
Disabled	Grey-100 fill, grey-disabled text, no border	Read-only mode fields (expired/revoked token)
Success	1px emerald border, small check icon inside field	Confirms client uniquely resolved (UC-06)

2.3 Status Badges

- SYNCED — Emerald pill, checkmark icon, label "Synced".
- PENDING — Amber pill, clock icon, label "Pending sync".
- BLOCKED — Red pill, lock icon, label "Blocked" (escrow ceiling or GPS gate).
- Booklet token — ACTIVE (emerald), EXPIRED (grey, "Read-only"), REVOKED (red, "Access

revoked"). **No GRACE state** — see §5.2.

3. Accessibility Specification

Requirement	Specification
Contrast	All text/background pairs meet WCAG 2.1 AA (4.5:1 body, 3:1 large text). Amber (#FFB800) always paired with black text, never white, to hold contrast in direct sunlight.
Touch targets	Minimum 48×48 dp for every interactive control on the Agent and Client apps (fat-finger spec); minimum 40×40 px on the web Back-Office.
Screen reader labels	Every icon-only control (SOS button, QR scan FAB, map markers) carries a semantic label — e.g. "Emergency SOS, activates after 3-second hold", "Scan client QR code".
Focus order	Collection wizard steps are traversed in strict DOM/semantic order 1 → 2 → 3; modals trap focus and return it to the triggering element on close.
Color independence	Status is never conveyed by color alone: sync/ceiling/geofence states pair a color with an icon and a text label (e.g. lock icon + "Ceiling reached", not just red fill).
Low-literacy support	Primary flows (client lookup, denomination entry, GPS gate) rely on icons, numerals and color-coded states over dense text, consistent with the CEMAC field-literacy constraint noted in the SAD design principles.
Motion sensitivity	The SOS long-press vibration and geofence-breach flash are capped to short, single-pulse patterns; no continuous flashing animations (seizure-safety).

4. Agent Mobile App (Flutter) — Collection Zone

4.1 Authentication (UC-01)

- Launch screen reads device IMEI silently in the background, then prompts PIN or biometric.

- Unknown/unbound IMEI: full-screen red block — "This device is not registered to an agent account. Contact your branch manager." No retry loop offered on-device. (*Verified: the backend rejects login with a distinct "Device IMEI does not match registered device" error whenever the submitted IMEI doesn't match the agent's registered one — this is enforced server-side, not just a UI convention.*)
- **3 failed PIN attempts locks the account for 15 minutes**, counted independently of IMEI/schedule failures; the API returns HTTP 423 with the exact unlock timestamp, so the app can render an accurate countdown. A correct PIN entered while still locked is still rejected — the lock is checked before the PIN itself. Both the attempt threshold and lockout duration are server-configurable, so treat "15 minutes" as the current default, not a hardcoded number to bake into the UI copy.
- A successful login clears the failed-attempt counter entirely, regardless of how many failed attempts preceded it.
- Offline authentication is supported against the encrypted local credential cache when no network is present.

4.2 Session Window & Working Hours (UC-15, FR-15, NFR-10)

- Outside the branch's configured open/close window, the app shows a locked dashboard state: "Collections are closed until [open_time]" with no wizard entry point.
- GPS background tracking (UC-10) automatically stops outside the working window, per NFR-10; the connectivity/tracking icon in the header reflects this as a neutral grey "Tracking paused" chip rather than the amber offline state.

4.3 "Fat-Finger" Accessibility Baseline

- Minimum action button height: 48 dp, for use in open-air markets with hands often full or gloved.
- Persistent top connectivity banner: hidden when online; amber "Offline Mode — [X] collections ready to sync" when offline; grey "Tracking paused" outside working hours.

4.4 Escrow Wallet & Ceiling (UC-03, UC-04, UC-05)

- Home dashboard shows a compact gauge for `escrow_account.balance_xaf` against `ceiling_xaf`, colored per the thresholds in Section 1.1.
- At 100% of ceiling, new collections are blocked at the client-lookup step with a red banner and lock icon, before the agent invests time entering client/denomination data.
- Remote ceiling overrides (UC-05, pushed from Back-Office) apply live without requiring an app restart, and display a countdown to `valid_until`.

4.5 Three-Step Collection Wizard (UC-06, UC-07, UC-08, UC-12)

[Step 1: Client] —(resolved)—> [Step 2: Denominations] —(GPS gate)—> [Step 3: Confirmation]

Step 1 — Client Resolution (UC-06)

- Search field supports QR scan, member ID, phone, or name; multiple matches present a disambiguation list rather than auto-selecting.
- On unique match: display client_profile card (name, member number, photo, branch). "Next" activates only once exactly one client is locked.

Step 2 — Denomination Breakdown (UC-08, BR-02)

- CEMAC Grid: one row per BEAC denomination (10000 / 5000 / 2000 / 1000 / 500 XAF) plus a coins-total field, integer-only quantities.
- Live running total = $\Sigma(\text{denomination} \times \text{quantity})$; "Next" is disabled until this total equals the amount the agent intends to collect (BR-02).

Step 3 — GPS Gate & Submission (UC-12, BR-05)

- On "Validate collection": query location services.
 - GPS active with valid fix: discreet loader, collection saved with lat/lon/accuracy/timestamp captured immutably (non-editable by the agent, FR-12).
 - GPS inactive or no fix: hard-blocking red pop-up — "Action impossible. GPS must be enabled to validate a collection." Submit stays disabled; this is a strict block, not a warning, per BR-05.
- Successful submission triggers UC-09 notification (SMS + optional Bluetooth print) and writes core.collection with sync_status = PENDING until synced.
- **The moment a collection reaches the server it is, by definition, already synced** — there is no "PENDING" row visible server-side. The app's own local queue (before a sync call succeeds) is the only place a truly pending collection exists. See §6.1 for how that local queue is made visible to the Back-Office.

4.6 Field Tracking & Emergency (UC-10, UC-11, UC-13, UC-14)

- Background GPS pings (~5 min cadence) are invisible to the agent except for the header tracking indicator; battery impact is minimized per NFR-09 (no dedicated screen).

- Geofence breach: agent-facing app does not need to surface the breach itself (this is a Back-Office concern) but should not alarm the agent — no pop-up interrupts their workflow. (*Verified: breach detection runs entirely server-side, evaluated inline on every location ping — there is no dedicated breach-notification endpoint the agent app could even call.*)
- SOS (UC-14): accessible via the side menu or a 3-second long-press on the app header. Triggers a single-pulse vibration and a discreet confirmation message; attempts SMS delivery even with no data connectivity. Cannot be disabled by the agent (FR-14).

5. Client Mobile App (Digital Booklet)

5.1 Activation & Renewal (UC-19)

Corrected in v3.0. The v2.0 draft described a single-app, agent-entered flow ("the client never enters the temporary Activation ID directly on their own device") backed by one API call. That is not what the backend implements, and was never meant to be: an early build of this exact flow was deliberately redesigned specifically because a single actor (the agent) being able to trigger the entire activation — commission payout included — is a fraud vector (an agent could activate a fictitious or uninformed client purely to collect their own commission). The real flow is a **two-party, two-device gate**: the client and the agent each take an independent, separately-authenticated action, and the booklet is only issued once *both* have happened.

Activation requires the Client app *and* the Agent app, used independently by each party:

- **Step 1 — Client self-activates, on their own device:** the client opens the Client app, enters the temporary Activation ID they received from the CBS, and sets their own permanent login and PIN. The app validates the Activation ID against the CBS before accepting it. This does not yet activate anything — it only establishes the client's credentials.
- **Step 2 — Agent sponsors, on their own device, in either order relative to Step 3:** the agent identifies the client by the login the client just chose (not by the Activation ID — the client never hands that over to the agent), and registers that they've received the activation fee in cash from the client, in person. This is real cash and counts against the agent's escrow ceiling exactly like an ordinary collection (BR-03).
- **Step 3 — Client confirms payment, on their own device:** the client, in their own authenticated session, re-enters their PIN to confirm the agent's cash-receipt record is correct.
- **Step 4 — Finalization:** the moment *both* Step 2 and Step 3 have happened (in either order), the system automatically splits the fee between agent commission and MFI share, and issues the 365-day access token. Neither party can finalize activation alone.
- **Stuck-gate handling:** if the other party never confirms, an administrator can cancel the pending gate (with a mandatory reason) from the Back-Office; a stale gate also auto-expires after a configurable

timeout (default 24h), freeing the agent to collect cash again in the meantime.

- Activation ID not found: the client's own app blocks credential-setting and redirects to physical branch registration; the app never creates a CBS customer record.

5.2 Booklet Token State Machine (UC-20/21/22 — `access_token.status`)

Corrected in v3.0. The backend's token model has **three** real states, not four — there is no persisted or computed GRACE state anywhere in the system today. "EXPIRED" is not a stored value either; it's computed on every read by comparing the token's expiry timestamp to the current time.

- **ACTIVE** — full interface; all monetary actions available ("Request withdrawal", renewal not yet needed). Persisted state.
- **EXPIRED** (read-only) — grey banner: "Annual token expired. Your data remains visible." Monetary action buttons drop to 30% opacity (inactive); balance and history stay sharp and fully legible (BR-04 — money is never locked by token status). *Computed, not stored:* the app should treat any token whose expiry timestamp has passed as EXPIRED, with no separate signal from the server.
- **REVOKED** — full-screen block distinct from EXPIRED: "Your digital access has been revoked. Please contact your branch." No balance or history is shown in this state, unlike EXPIRED. Persisted state.

If a GRACE-period renewal reminder is still wanted (a non-blocking "renew soon" banner shown for some window before hard expiry), that is a real, buildable feature — it just doesn't exist yet. It would need a configured reminder-window value and a third computed state alongside ACTIVE/EXPIRED, mirroring how EXPIRED itself is computed today. Flag this to the product owner as a scoping decision, not an implementation detail to assume.

5.3 Home / Balance (UC-21)

- Home screen visually echoes the traditional cardboard savings passbook (textured beige or brand-navy background).
- Balance is shown in 32 pt bold regardless of token state (except REVOKED); falls back to last-cached balance when the CBS is unreachable, with a small "as of [timestamp]" caption. Balance access is not gated by token status at all — it works the same whether the token is ACTIVE or EXPIRED.

5.4 Contribution History / Mini-Statement (UC-22)

- Filterable, append-only list of collections tied to the client; supports an offline cached view.
- Empty state and USSD short-code equivalent are covered in Section 8.

6. End-of-Day Cash Closeout Screens (OFJ)

6.1 Reconciliation Interface (UC-16, BR-01)

Field	Value
TOTAL DIGITAL (core.ofj_agent_line.digital_total_xaf)	450,000 XAF
TOTAL PHYSICAL (cashier-entered, ofj_physical_denom)	445,000 XAF
Δ VARIANCE (physical – digital, BR-01)	– 5,000 XAF

- "Close day" is disabled — not merely unhighlighted — while any agent in the branch is still reconciling, **or while any agent currently reports a nonzero local sync queue**, regardless of the Δ value on either count.
- Δ = physical – digital, computed live as the cashier enters ofj_physical_denom quantities.
- $\Delta == 0$ (and no agent has a pending queue): "Close day" activates in Emerald. $\Delta != 0$: button stays locked and the variance modal opens automatically.

Refined in v3.0. A collection genuinely queued on an agent's phone, not yet synced, has no server-side representation at all — the server has no way to know about a transaction it has never received, so there is no literal `core.collection.sync_status = PENDING` row to check. Instead, the Agent app must periodically self-report its local queue depth via `PATCH /agents/{id}/sync-status` (body: `{"pendingCount": N}`) — the same kind of lightweight background self-report the app already does for GPS pings. OFJ close is blocked server-side while any active agent in the branch currently reports a nonzero count, independent of how the reconciled lines look. Recommended cadence: report on every successful/failed sync attempt, and once more when the connectivity banner (§4.3) changes state.

6.2 Cash Variance / Agent Debt Modal (UC-17)

- Orange-headed warning modal.
 - Required justification field — confirm stays inactive under 10 characters.
 - Confirm action: "Confirm transfer of the variance to the agent's debt account" — writes a `variance_debt` row; this action is restricted to $\Delta < 0$ (shortages only, per FR-17); a $\Delta > 0$ surplus is annotated but not turned into a debt.

6.3 Daily CBS Export (UC-18)

- Export action is available only after the OFJ session is closed (status = CLOSED).
- Export screen shows format, generated_at, and ack_status; a mapping error surfaces inline with a retry action rather than silently failing.

7. Web Management Portal (Next.js 16) — Back-Office

7.1 Agent Grid & Wallet Ceiling Alerts (UC-04)

- Compact list: Name, Zone, Escrow Wallet Level, with the same gauge thresholds as the mobile app for visual consistency.
- 100% ceiling shows a lock icon next to the agent's name, signalling their field collections are automatically blocked.

7.2 Ceiling Waiver Modal (UC-05)

- Fields: temp_ceiling_xaf, valid_until (date/time picker), and a required "Reason for waiver" text field.
- Requires an approver role (approved_by) per the ceiling_override schema; the override auto-restores the original ceiling at valid_until.

7.3 Agent Enrollment & Lifecycle (UC-02)

- Admin form: identity, branch, escrow reference, IMEI — duplicate IMEI is rejected inline.
- Suspend action invalidates all active sessions immediately; deletion is blocked while sessions are open or a variance_debt is unresolved (FR-02).

7.4 Branch Working-Hours Configuration (UC-15)

- Simple open/close/grace time picker per branch, with a history log of prior schedule changes retained for audit.

7.5 Mapping & Field Tracking Module (UC-11, UC-13)

- Full-screen Mapbox/Leaflet component.
 - Route trace: polyline connecting the day's location_ping records.
 - Transaction markers: clickable points opening the digital receipt.

- Geofencing: translucent polygon per assigned geofence; a ping outside the polygon turns the route line red instantly and raises an entry in the notifications panel (geofence_alert).

7.6 OFJ & Variance Oversight

- Branch-scoped table of open/closed OFJ sessions, outstanding variance_debt, and export ack_status — mirrors the mobile OFJ data but read/oversight-only from this surface.

7.7 Emergency / SOS Console

- Live-updating list of raised SOS events, most recent first, with an acknowledge action once a manager has responded. Acknowledged events remain visible in the list (never deleted) for audit, distinguishable by an "Acknowledged" badge instead of the flashing red "Active" state.
- Branch-scoped for Branch Manager/Cashier roles; an org-wide ADMIN sees every branch's alerts in one list.

8. Error, Loading & Empty States

8.1 Network Error Dialogs (Flutter)

In a context of intermittent connectivity, the UI must never alarm the agent or client. It must clearly distinguish a connectivity issue from a payment failure.

	Error A — Sync Failure	Error B — Core Banking Timeout
Trigger	Background sync call (POST /collections/sync) fails — agent is offline	Agent has connectivity, but Amplitude/FinanSoft does not respond within SLA
Component	Persistent toast / top banner (Snackbar)	Blocking modal, shown on collection submission
Color	Amber #FFB800, black text	Navy header #0A2540, discreet spinner
Exact UI copy	" Offline Mode enabled. Your collections are safely stored on the phone. They will be sent as soon as the	" Record saved successfully. The collection has been recorded. The client's official bank balance update is being processed (bank connection

	network returns. (Tickets pending: X)"	temporarily slow)."
Action / icon	Standard network icon replaced by a database icon with a small lock (AES-256 offline store, NFR-01)	Single 48 dp button: [Continue my collections] in Emerald #00D4B2
Underlying mechanism	core.collection row persists locally with sync_status = PENDING; unique (agent_id, device_tx_id) prevents duplicate posts on retry	A direct synchronous call from Core to the Middleware , guarded by an idempotency key (mw.idempotency_key) so a retried export/post never double-posts once the CBS recovers. <i>See correction below.</i>

Corrected in v3.0. The v2.0 draft attributed this resilience to "RabbitMQ decouples Core from CBS." That queue exists in the stack, but the actual Core → Middleware call for both the daily export and individual collection posting is a **direct synchronous HTTP call** (Core's CbsClientService, via WebClient), not something routed through RabbitMQ. The non-blocking guarantee the UI copy above promises comes from two other real mechanisms instead: (1) the collection itself is already saved and visible to the agent before this call is even attempted, so a slow/failed CBS call never blocks the agent's next action; and (2) the idempotency key means a retried call (whether retried by the app, by an admin re-running export, or automatically) is safe. If true queue-based decoupling is wanted here, it does not exist yet and would be new work, not a UI-copy fix.

8.2 Loading & Empty States by Screen

Screen	Loading state	Empty state
Client lookup (UC-06)	Skeleton row placeholders while querying by QR/ID/phone/name	"No client found — try another search method or register offline."
Escrow dashboard (UC-04)	Gauge shimmer while balance/ceiling loads	Not applicable — every agent has an escrow_account
Route map (UC-11)	Grey polyline skeleton while pings load	"No location pings recorded for this date."
Contribution history	3 skeleton rows	"No contributions yet — your history will

(UC-22)		appear here after your first deposit."
OFJ reconciliation (UC-16)	Spinner on digital total while pending syncs finish	Blocked state: "Reconciliation unavailable — pending offline collections must sync first."
Geofence alerts panel	Spinner on initial load	"No active breaches — all agents within assigned zones."
SOS console	Spinner on initial load	"No SOS events — nothing to review."

9. Motion & Interaction Specification

- Standard transition: 150–200 ms ease-out for screen pushes and modal entry; no motion longer than 300 ms anywhere in the agent app (speed matters more than flourish in the field).
- SOS long-press: a single, short vibration pulse (~150 ms) confirms activation — no repeating buzz pattern.
- Geofence breach: the route line color transition is instantaneous (no animation) since it functions as an alert, not decoration.
- Loading spinners are discreet (16–20 px, single color) and never block the whole screen except for the two modal cases defined in Section 8.1.
- Reduced-motion: respect the OS-level "reduce motion" accessibility setting by disabling non-essential transitions (card entrance animations) while preserving state-change indicators (spinners, badges).

10. Localization & Content Strategy

- Primary languages: French (default, CEMAC francophone zones) and English, selectable at first launch and in Settings; the Back-Office mirrors both.
- All UI copy lives in externalized string tables (ARB for Flutter, i18n JSON for Next.js) — no hardcoded strings in the wizard, error dialogs, or receipt template.
- Currency and number formatting always use the space-separated thousands rule (Section 1.2) in both locales; date/time follow ISO-derived local formats, stored in UTC (TIMESTAMP TZ) and rendered in branch-local time (branch.timezone).
- Text expansion budget: French strings run ~15–20% longer than English; buttons and banners must accommodate this without truncation, especially the 48 dp action buttons.
- Low-literacy support: icons and color coding (Section 1.1, 3) are treated as a first-class communication

channel alongside text, not a decoration, consistent with the SAD's low-literacy field-user constraint.

11. Bluetooth Thermal Receipt Template (UC-09)

Printed via a portable Bluetooth thermal printer (58 mm width, monospace font) — the most effective way to build trust with merchants and savers in the field.

```
=====
                MICROFI
        DIGITAL CASH COLLECTION CEMAC
=====
Branch : Yaounde - Central Market
Date   : 03/08/2026 09:30 UTC
Agent  : AF-042 (Fouda M.)
-----

CLIENT ID : MFI-2026-88391
NAME      : KAMGA ADELE
-----

AMOUNT COLLECTED : 25 000 XAF
(Twenty-Five Thousand CFA Francs)

Payment Method : CASH
Ticket Status  : RECORDED (*)

-----
                DENOMINATION BREAKDOWN (XAF)
-----
10 000 x 2 = 20 000
 5 000 x 1 =  5 000
 2 000 x 0 =      0
 1 000 x 0 =      0
   500 x 0 =      0
-----

UNIQUE REF : device_tx_id (local UUID)
SECURE DIGITAL SIGNATURE:
[XAF-GEOTAG-VALID-030826]
-----

(*) This electronic receipt serves as proof
of your deposit in case of temporary
network outage.
```

=====

THANK YOU FOR YOUR TRUST

=====

- Denomination lines print only if quantity > 0, to conserve thermal paper in the field.
- UNIQUE REF shows device_tx_id, generated locally, enabling immutable transaction traceability for COBAC/BEAC even if sync happens hours later.
- Alignment: automatic centering for header blocks; right alignment for financial amounts.

12. Asset Export Specification

- Icons: single SVG source per icon, exported at 24×24 and 32×32 for mobile (@1x/2x/3x raster fallback), 20×20/24×24 for web; stroke-based, 2 px stroke weight, consistent corner radius with the design-token radius scale.
- Illustrations (empty states, onboarding): SVG, optimized (<15 KB), no embedded raster unless photographic (client profile photo placeholder).
- Naming convention: icon/{name}/{size}, illustration/{context}/{name} — matches component/token naming so engineering can map 1:1 without a lookup table.
- Map assets (geofence polygons, markers): exported as GeoJSON, not images, to stay editable in the Mapbox/Leaflet layer.

13. Acceptance Criteria (Definition of Done)

Each screen below must satisfy its checklist before engineering marks the corresponding UC as UI-complete.

Screen	Definition of Done (sign-off checklist)
Collection Wizard (UC-06/07/08/12)	☐ Client resolves to exactly one profile before Step 2 unlocks · ☐ Denomination sum auto-updates on every keystroke and blocks submit on mismatch (BR-02) · ☐ Submit is disabled with red modal when GPS has no valid fix (BR-05) · ☐ Successful submit shows Emerald confirmation and queues SMS/print (UC-09) · ☐ Works fully offline with encrypted local write (NFR-01).
Escrow Gauge (UC-04/05)	☐ Gauge color matches balance/ceiling ratio thresholds exactly (<80 / ≥80 / 100%) · ☐ Lock icon appears only at 100% · ☐ Override modal cannot submit without a reason and a

	valid_until date-time · ☐ Override auto-expires and reverts the ceiling.
OFJ Reconciliation (UC-16/17)	☐ Close button is disabled while any collection has sync_status = PENDING or any agent reports a nonzero local sync queue (PATCH /agents/{id}/sync-status) · ☐ Δ = physical – digital computed live · ☐ Δ ≠ 0 auto-opens the variance modal · ☐ Variance modal blocks confirm under 10 characters of justification · ☐ Confirmed variance creates a variance_debt row and unblocks close.
Client Digital Booklet (UC-19– 22)	☐ Client sets credentials on their own device via the Activation ID, independent of the agent · ☐ Agent sponsorship and client payment confirmation are two separate, independently-authenticated actions, in either order · ☐ Token/fee-split finalizes only once both are done · ☐ ACTIVE state shows full actions · ☐ EXPIRED state disables monetary actions at 30% opacity while balance stays sharp · ☐ REVOKED state fully locks the app to a support/contact message, distinct from EXPIRED · ☐ Balance and history remain visible in every non-REVOKED state (BR-04).
Geofencing & SOS (UC-13/14)	☐ Breach turns the live route line red within one ping cycle · ☐ Alert is logged and shown in the Back-Office notification panel · ☐ SOS triggers on a 3-second long-press only, never on a single tap · ☐ SOS attempts SMS even with no data connectivity.
Agent Login (UC-01)	☐ Unbound IMEI blocks login with a distinct, non-retryable message · ☐ 3rd consecutive failed PIN attempt returns HTTP 423 with an exact unlock timestamp, before any further PIN check · ☐ A successful login clears the failed-attempt count · ☐ Suspended-account and outside-schedule-window rejections are visually distinct from a wrong PIN.

14. Traceability Matrix — UC → Screen → API → Entity

UC	Screen / Interaction	API endpoint	Primary entity
UC-01	Agent login (IMEI + PIN/biometric)	POST /auth/agent/login	core.agent
UC-02	Agent enrollment / suspend (Back-Office)	GET/POST /admin/agents, PATCH .../status	core.agent

UC-03	Escrow top-up	POST /agents/{id}/escrow/top-up	core.escrow_ledger
UC-04	Escrow ceiling gauge & lockout	GET /agents/{id}/escrow	core.escrow_account
UC-05	Temporary ceiling override modal	PATCH /admin/agents/{id}/ceiling	core.ceiling_override
UC-06	Multi-channel client lookup	GET /clients/lookup	core.client_profile
UC-07	Offline collection wizard step 2/3	POST /collections, POST /collections/sync	core.collection
UC-08	Denomination breakdown grid	POST /collections (nested)	core.denomination_line
UC-09	Post-validation SMS/receipt	POST /collections/{id}/notify	core.notification_log
UC-10	Periodic GPS ping (background)	POST /agents/{id}/location	core.location_ping
UC-11	Route visualization map	GET /admin/agents/{id}/route	core.location_ping
UC-12	GPS gate + geotag on submit	POST /collections	core.collection (lat/lon/accuracy)
UC-13	Geofence breach alert	n/a (server-evaluated on ping ingest); alerts listable via GET /admin/agents/{id}/geofence-alerts	core.geofence, core.geofence_alert
UC-	SOS emergency button	POST /agents/{id}/sos	core.sos_event

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UC-15	Branch working-hours config	GET/PUT /admin/branches/{id}/schedule	core.branch
UC-16	OFJ reconciliation screen	GET /ofj/{branchId}/summary, POST .../reconcile	core.ofj_session, ofj_agent_line
UC-17	Cash variance → agent debt modal	POST /ofj/{branchId}/variance	core.variance_debt
UC-18	Daily CBS export	POST /ofj/{branchId}/export	core.export_batch
UC-19	Client credential self-activation, on the client's own device	POST /auth/client/activate	core.client_profile
UC-19	Agent sponsorship (identifies client by login)	POST /clients/activation	core.activation_request, activation_payment
UC-19	Client payment confirmation, on the client's own device	POST /clients/me/activation/pay	core.access_token, activation_payment
UC-20	Client profile self-service	GET /clients/me/profile	core.client_profile
UC-21	Balance & summary (token-independent)	GET /clients/me/balance	core.client_profile / CBS
UC-22	Contribution history / mini-statement	GET /clients/me/history	core.collection (client-scoped)

14.1 Back-Office Operational Endpoints (beyond the UC-01–22 matrix)

The following support the Back-Office screens in Section 7 and are not modeled as distinct Business Analysis

use cases, but are required for those screens to function:

Screen / Interaction	API endpoint
Local sync-queue self-report (feeds §6.1's OFJ close guard)	PATCH /agents/{id}/sync-status
SOS console listing / acknowledge (§7.7)	GET /admin/sos-events, PATCH .../{id}/acknowledge
OFJ history (past business days, not just today)	GET /ofj/{branchId}/history
Variance debt dashboards (branch-wide and per-agent)	GET /ofj/{branchId}/variance-debts, GET /admin/agents/{id}/variance-debts
Notification delivery audit (was this SMS actually sent)	GET /collections/{id}/notifications
Client/agent/admin-user detail & directory screens	GET /admin/clients, GET /admin/clients/{id}, PATCH .../status, GET /admin/agents/{id}, GET/PATCH /admin/users/{id}[role/password]

15. Glossary

Term	Definition
ACID	Atomicity, Consistency, Isolation, Durability — transactional guarantees used for money-moving paths (Spring Data JPA).
AES-256	Advanced Encryption Standard, 256-bit key — encrypts the offline collection queue on the agent's device (NFR-01).
AMQP	Advanced Message Queuing Protocol — the messaging protocol used by RabbitMQ. In the current backend this decouples auth-event audit logging, not the CBS export path

	(see §8.1).
BEAC	Banque des États de l'Afrique Centrale — the central bank issuing CEMAC banknotes; defines the denomination set (500–10,000 XAF).
BR	Business Rule — a numbered rule from the Business Analysis (e.g. BR-05: GPS is mandatory before any collection).
Back-Office	The Next.js 16 web application used by branch managers and administrators.
CBS	Core Banking System — the bank's system of record (e.g. Amplitude, FinanSoft) that MICROFI integrates with via the Middleware.
CEMAC	Communauté Économique et Monétaire de l'Afrique Centrale — the six-country monetary zone MICROFI operates in.
COBAC	Commission Bancaire de l'Afrique Centrale — the regional banking regulator requiring transaction traceability.
Ceiling (escrow ceiling)	The maximum cash an agent may hold before collections are automatically blocked (BR-03).
Denomination	A single banknote or coin face value (e.g. 1,000 XAF); collections are broken down by denomination and quantity (FR-08).
Device TX ID (device_tx_id)	A locally generated unique transaction identifier that lets a collection be traced even if sync happens hours later.
Escrow account	The agent's cash guarantee wallet (escrow_account.balance_xaf against ceiling_xaf) that limits field cash exposure.
FR	Functional Requirement — a numbered requirement from the Business Analysis (e.g. FR-12: mandatory GPS geotagging).
Geofence	A drawn geographic boundary an agent is expected to stay within; breaches raise a

	geofence_alert.
IMEI binding	Locking an agent's session to a single physical device's IMEI, part of authentication security (FR-01).
JWT	JSON Web Token — the signed token format used for agent, client and admin session authentication.
MVP	Minimum Viable Product — the reduced feature set targeted for the initial 6-week production window.
NFR	Non-Functional Requirement — a numbered quality attribute (e.g. NFR-09: battery-usage optimisation for GPS tracking).
OFJ	Opération de Fin de Journée (End-of-Day operation) — the daily cash-reconciliation and closeout process.
PIN lockout	Server-enforced cooldown after repeated failed PIN attempts (default: 3 attempts, 15 minutes, both server-configurable). Returns HTTP 423 with the unlock timestamp.
Pending sync count	An agent's self-reported count of collections queued locally but not yet synced (PATCH /agents/{id}/sync-status); blocks OFJ close while nonzero.
RabbitMQ	The message broker used in this stack for auth-event audit logging. Not currently on the Core → Middleware CBS export path — that call is direct and synchronous (see §8.1).
Sync status (sync_status)	The state of a collection record: PENDING (not yet sent) or SYNCED (confirmed received by the server). A record only ever exists server-side once SYNCED — see §6.1.
STRIDE	A threat-modeling framework — Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege.
UC	Use Case — a numbered behavioural specification (e.g. UC-16: OFJ reconciliation) that

	this handoff's screens map to.
Variance (Δ)	The difference between the physically counted cash and the digitally recorded total at end-of-day (BR-01).
Variance debt	A formal debt recorded against an agent when a cash shortage (negative variance) is confirmed (UC-17).
XAF / FCFA	The CFA franc (BEAC), the currency used across the CEMAC zone; displayed with a non-breaking space before the code, e.g. "45 000 XAF".